

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA14

COURSE NAME: Compiler Design

COURSE OUTCOME COVERED

CO4: Examine intermediate code generation techniques to enhance code optimization (BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 4

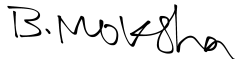
Total Marks:40

Annexure A

CLASS TEST

Code of Conduct:

I_B.MOKSHA SREE (192511160)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Q. No	Understanding of Concepts (25)	Explanation (25)	Application (20)	Organization(15)	Accuracy(10)	Total Score (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	
OVERALL RUBRICS VALUE						
	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 25	/ 30	/ 20	/ 15	/ 10	/ 100

Annexure A

CLASS TEST

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	15%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps

Name:- B. Moksha sree

Reg No:- 192511160

Course:- Compiler design

Course code:- CSE41402

"Cou-At1"

Class test

(Q1) If $((a < b) \text{ and } (c < d) \text{ or } (a > d))$ then

$z = x + y * z$

else

$z = z + 1$

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(A) if $((a < b) \text{ and } (c < d) \text{ or } (a > d))$

$z = x + y * z$

else

$z = z + 1$

three address code for the statement

L1:- if $a < b$ goto L2
goto L5

L2:- if $c < d$ goto L4
goto L3

L3:- if $a > d$ goto L4
goto L5

L4:- $t1 = y * z$
 $t2 = x + t1$
 $z = t2$
goto L6

L5:- $z = z + 1$

L6:- $t3 = z + 1$
 $z = t3$

2) Translate $(a < b)$ and $(c < d)$ or $(e < f)$ into three address statements using back patching.

$[(a < b) \text{ and } (c < d)] \text{ or } (e < f)$

1) if $(a < b)$ goto 3

2) goto 5

3) if $(c < d)$ goto 7

4) goto 5

5) if $(e < f)$ goto 7

6) goto 5

7) True

8) False.

(3) $A = (-A+B) * ((-A+B)*C)$

$t_1 = -A$

$t_2 = t_1 + B$

$t_3 = -A$

$t_4 = t_3 + B$

$t_5 = t_4 * C$

$t_6 = t_2 * t_5$

$A = t_6$

"three address code"

for the Expression

$A = (-A+B) * ((-A+B)*C)$

OP	arg1	arg2	result
Unary	A	-	t1
+	t1	B	t2
Unary	A	-	t3
+	t3	B	t4
*	t4	C	t5
*	t2	t5	t6
=	A6	-	A6

(Q4) Construct three address code using the following Production for Boolean Expression. $(a > b) \text{ or } (c < d) \text{ or } (e < f)$

$B_1 \rightarrow B_1 || B_2$

$B \rightarrow B_1 \&\& B_2$

$B \rightarrow ! B_1$

$B \rightarrow E_1 \text{ rel } E_2$

$B \rightarrow \text{true}$

$B \rightarrow \text{false}$

U:- $\text{if } (a > b) \text{ goto LTRUE}$
 go to ~~END~~ L2

L2:- $\text{if } (c < d) \text{ goto LTRUE}$
 go to L3

L3:- $\text{if } (e < f) \text{ goto LTRUE}$
 go to LFalse.

rect)

L TRUE:-

// Boolean expression is true
goto LEND

L FALSE:-

// Boolean expression is false

LEND)